



Chapter 2.2

Fuzzy Reasoning



Fuzzy Logic Controller

- Fuzzy Logic Controller (FLC) was first developed by Mamdani and Assilian around 1975 [1]
- The performance of an FLC depends on its Knowledge Base (KB)
- KB consists of both Data Base (DB) (that is, data related to membership function distributions of variables of the process to be controlled) as well as Rule Base (RB).
- Designing a proper KB can be done by the following manner
 - Optimization of DB only
 - Optimization of RB only
 - Optimization of DB and RB in stages
 - Optimization of DB and RB in simultaneously

Fuzzy Logic Controller (Cont.)



- A rule
IF A Student is **HardWorking** **AND** He is **Intelligent** **THEN** Result will be **Good**.
- **HardWorking, Intelligent, Good** these are fuzzy terms.
- Number of rules to be present in the rule base will increase, as the number of linguistic terms used to represent the variables increases.
- Computational complexity of the controller will increase with the number of rules.
- An FLC does not have an internal optimization module.
- An external optimizer is to be used to develop its optimal KB.

Steps involved in the working cycle of FLC



- The **condition** (also known as **antecedent**) and **action** (also called **consequent**) variables needed to control a process are identified and measurements are taken of all the condition variables.
- Measurements taken in the previous step are converted into appropriate fuzzy sets to express measurement uncertainties. This process is known as **fuzzification**.
- The fuzzified measurements are then used by inference engine to evaluate the control rules stored in the fuzzy rule base and a fuzzified output is determined.
- The fuzzified output is then converted into a single crisp value. This conversion is, called **de-fuzzification**. The de-fuzzified values represent the actions to be taken by the FLC in controlling the process.

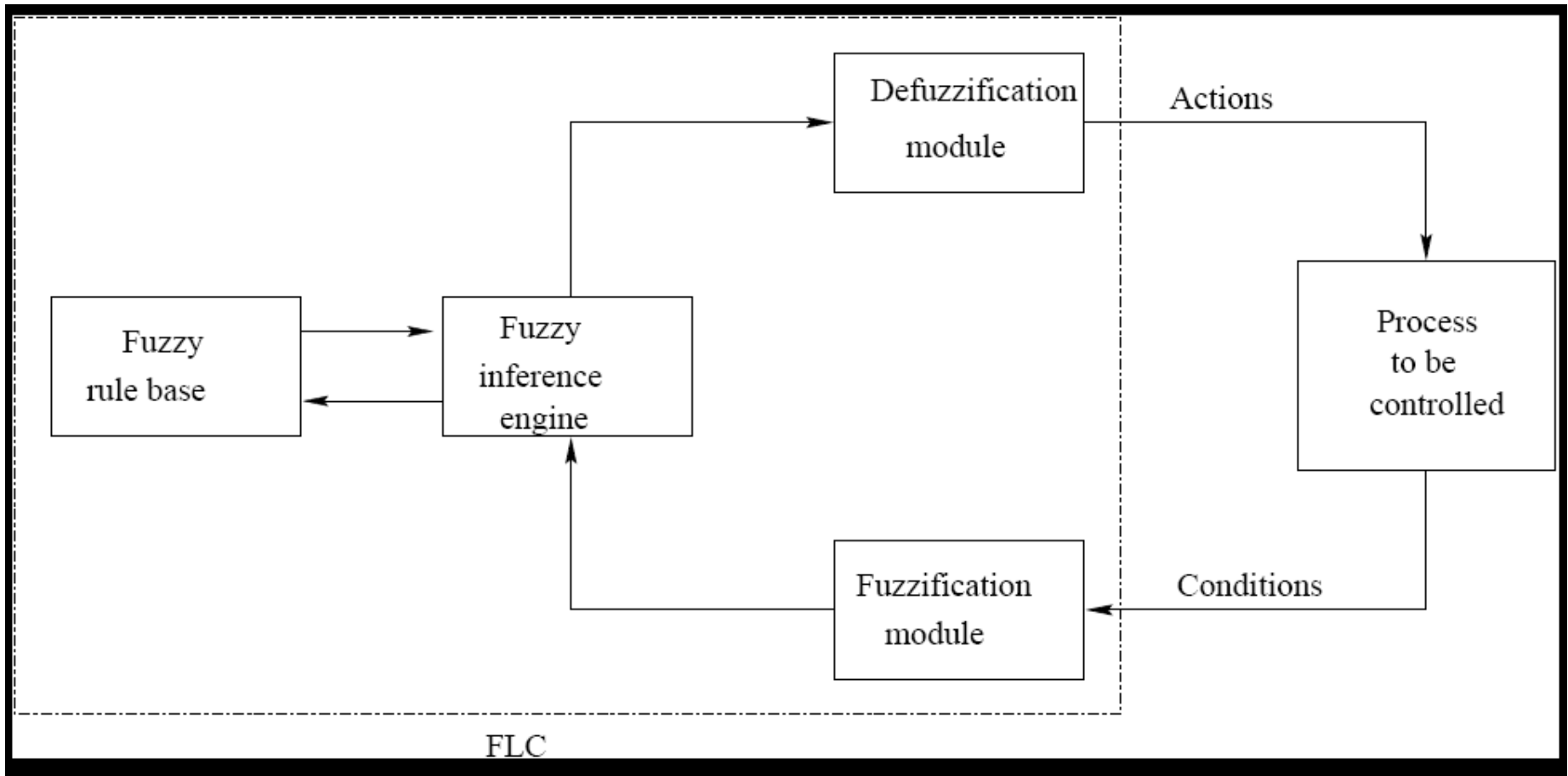


Two Major Forms of Fuzzy Logic Controller

- **Linguistic Fuzzy Modeling:** High interpretability and low accuracy.
Ex: Mamdani Approach[1]
- **Precise Fuzzy Modeling:** Low interpretability and high accuracy.
Ex: Takagi and Sugeno's Approach[2]



Mamdani Approach

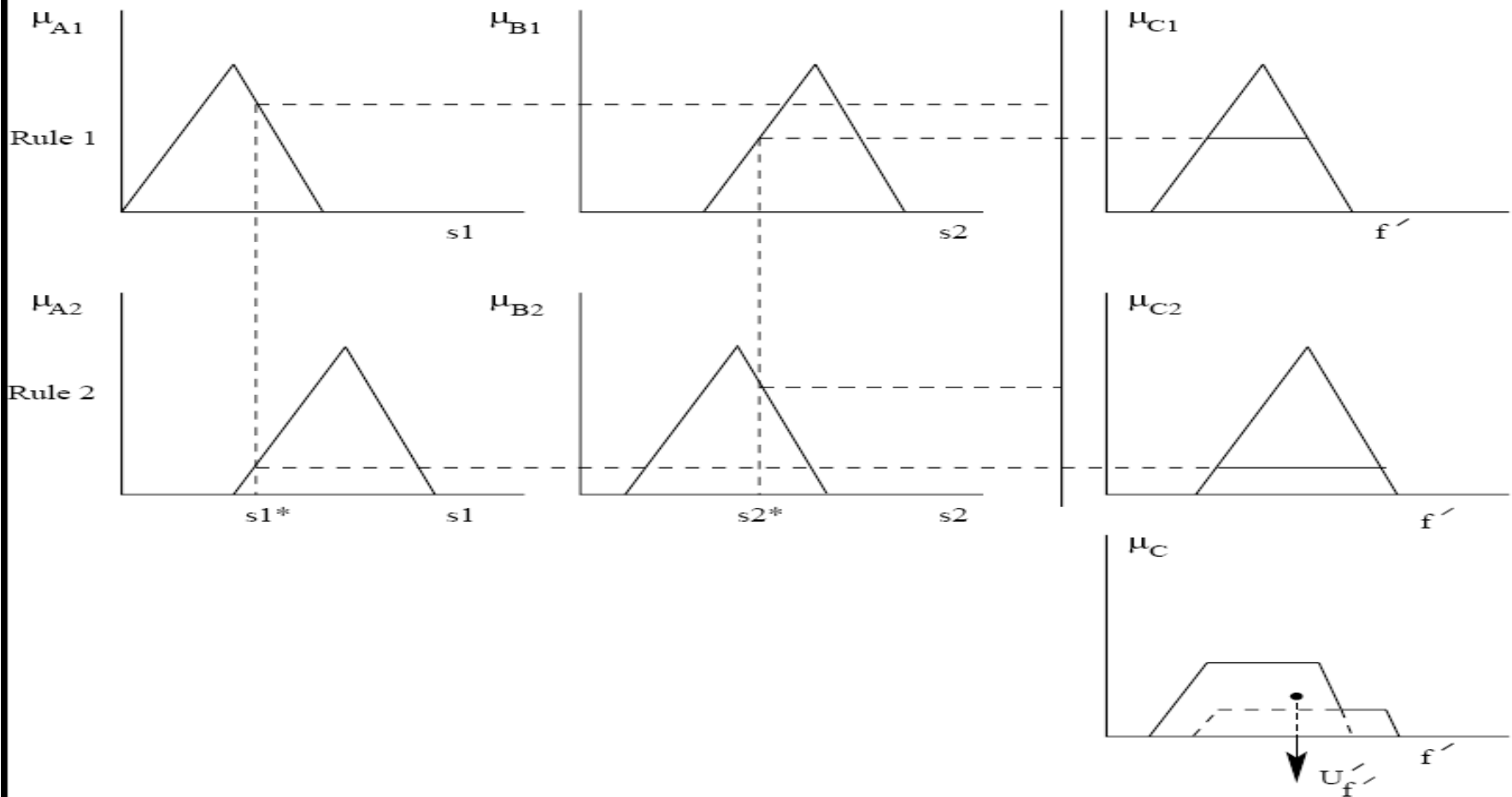




Mamdani Approach (Cont.)

Rule1: IF S_1 is A_1 AND S_2 is B_1 THEN f is C_1

Rule2: IF S_1 is A_2 AND S_2 is B_2 THEN f is C_2





Mamdani Approach (Cont.)

Let S_1^* and S_2^* are the inputs for fuzzy variables S_1 and S_2

Firing strengths of the first and second rules can be calculated as follows:

$$\alpha_1 = \min(\mu_{A_1}(S_1^*), \mu_{B_1}(S_2^*))$$

$$\alpha_2 = \min(\mu_{A_2}(S_1^*), \mu_{B_2}(S_2^*))$$

Membership value of the combined control action C is given by

$$\mu_C(f') = \max(\mu_{C_1}^*(f'), \mu_{C_2}^*(f'))$$



Methods of defuzzification

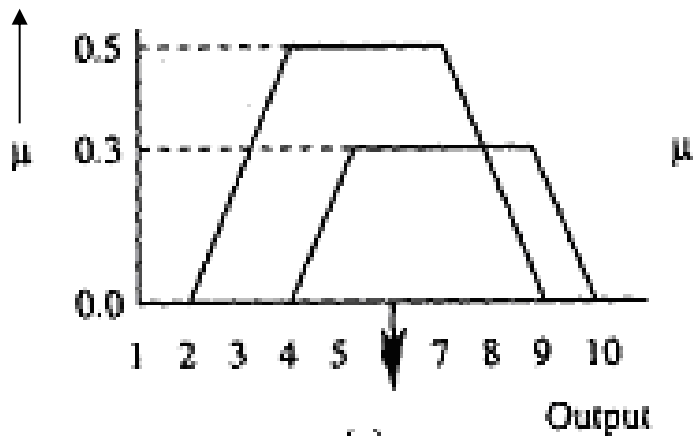
- Center of Sums Method
- Centroid Method
- Mean of Maxima Method

Center of Sums Method



where $A(\alpha_j)$: firing area of j-th rule
 f_j : center of the area
 p : No. of fired rules

$$U'_{f'} = \frac{\sum_{j=1}^p A(\alpha_j) * f_j}{\sum_{j=1}^p A(\alpha_j)},$$



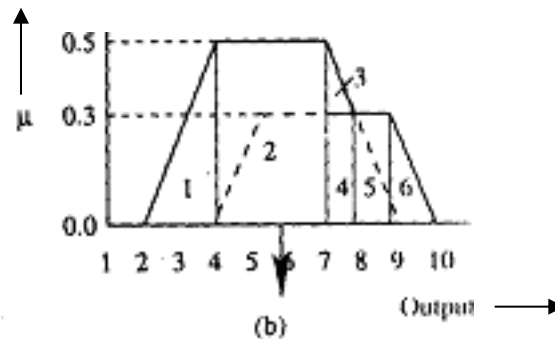


Centroid Method

Crisp output of the controller can be calculated using the expression

$$U'_{f'} = \sum_{i=1}^N A_i f_i / \sum_{i=1}^N A_i$$

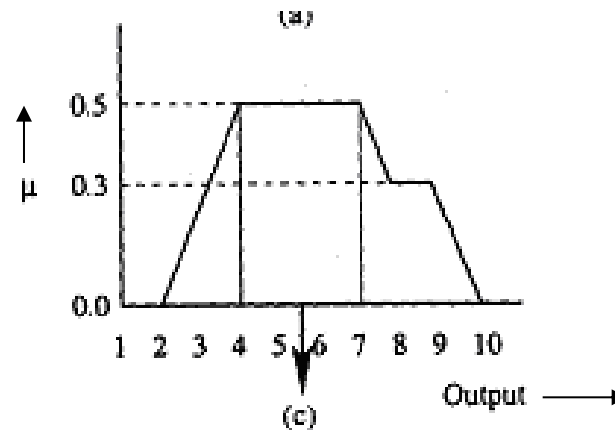
Where N indicates the number of small areas or regions, A_i and f_i represent the area and center of area, respectively, of i-th small region.





Mean of Maxima Method

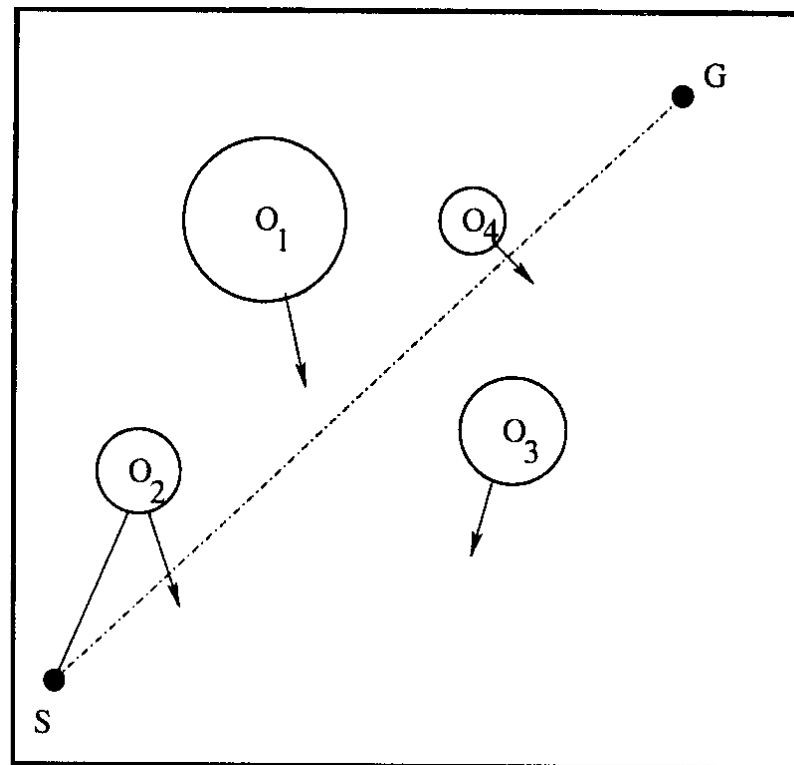
From the membership function distribution of combined control action, the range of the output variable corresponding to the maximum value of membership is identified. The mid-value of this range is considered to be the crisp output of the controller.





A Numerical Example

Figure shows a typical problem scenario related to navigation of a mobile robot in the presence of four moving obstacles.





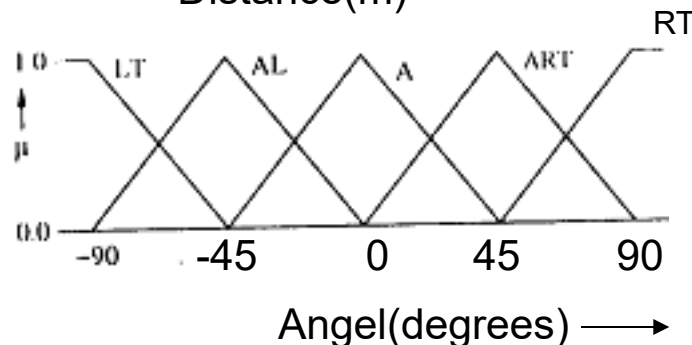
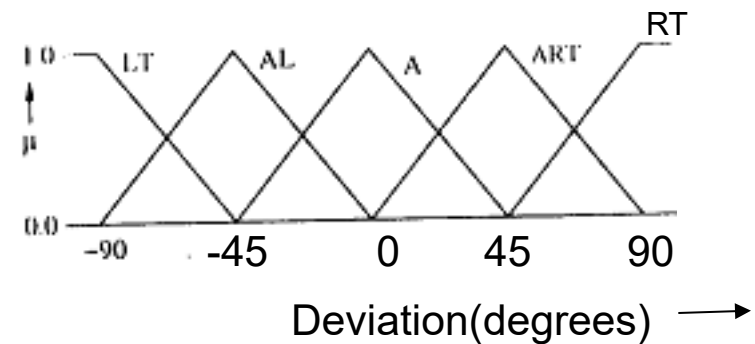
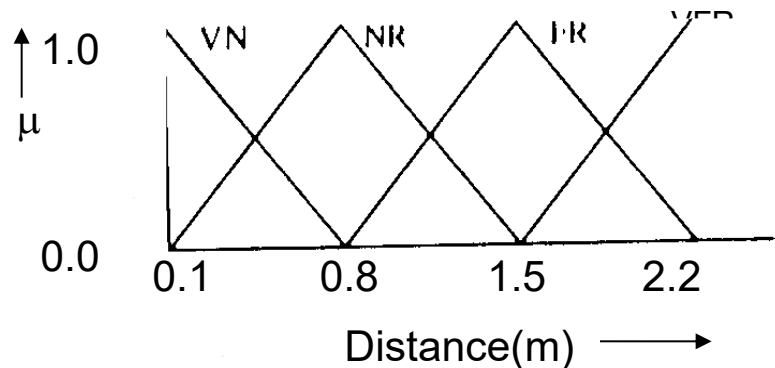
A Numerical Example (Cont.)

- Develop a fuzzy logic-based motion planner that will be able to generate the collision-free path for the robot.
- Inputs
 - Distance between the robot and the obstacle (D)
 - Angle (θ)
- Output
 - Deviation



A Numerical Example (Cont.)

- Distance is represented using four linguistic terms, namely Very Near (VN), Near (NR), Far (FR) and Very Far (VFR)
- Input: angle and output: deviation are expressed with the help of five linguistic terms, such as Ahead Left (AL), Left (LT), Ahead Right (ART) and Right (RT).





A Numerical Example (Cont.)

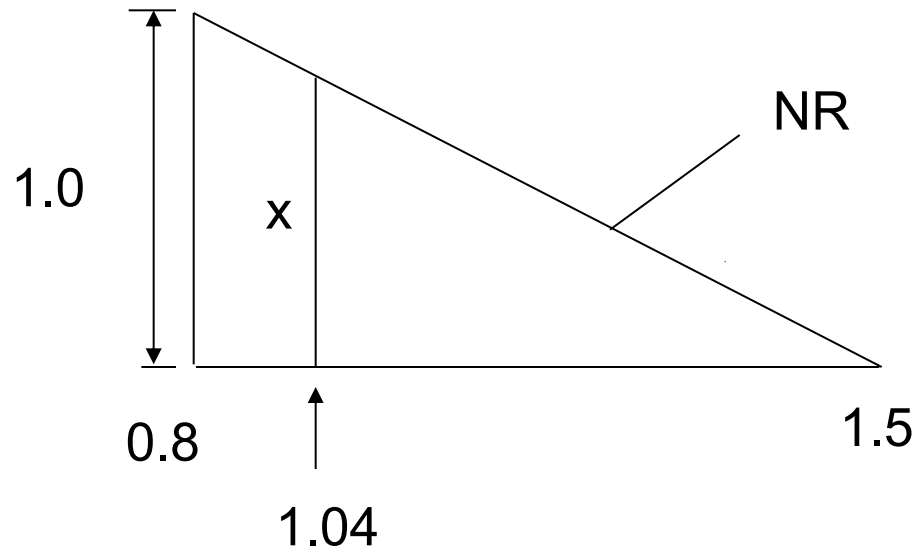
Rule Base(RB) of the Fuzzy Logic Controller (FLC)

Distance	Angel					
		LT	AL	A	ART	RT
	VN	A	ART	AL	AL	A
	NR	A	A	RT	A	A
	FR	A	A	ART	A	A
	VFR	A	A	A	A	A



A Numerical Example (Cont.)

- Determine the output deviation for the set of inputs: distance $D=1.04\text{m}$ and angle $GSO_2=30$ degrees
- The distance of 1.04 m may be called either NR (Near) or FR (Far)
- $\mu/1.0=(1.5-1.04)/(1.5-0.8)$ or $\mu_{NR}=0.6571$
- $\mu_{FR}=0.3429$



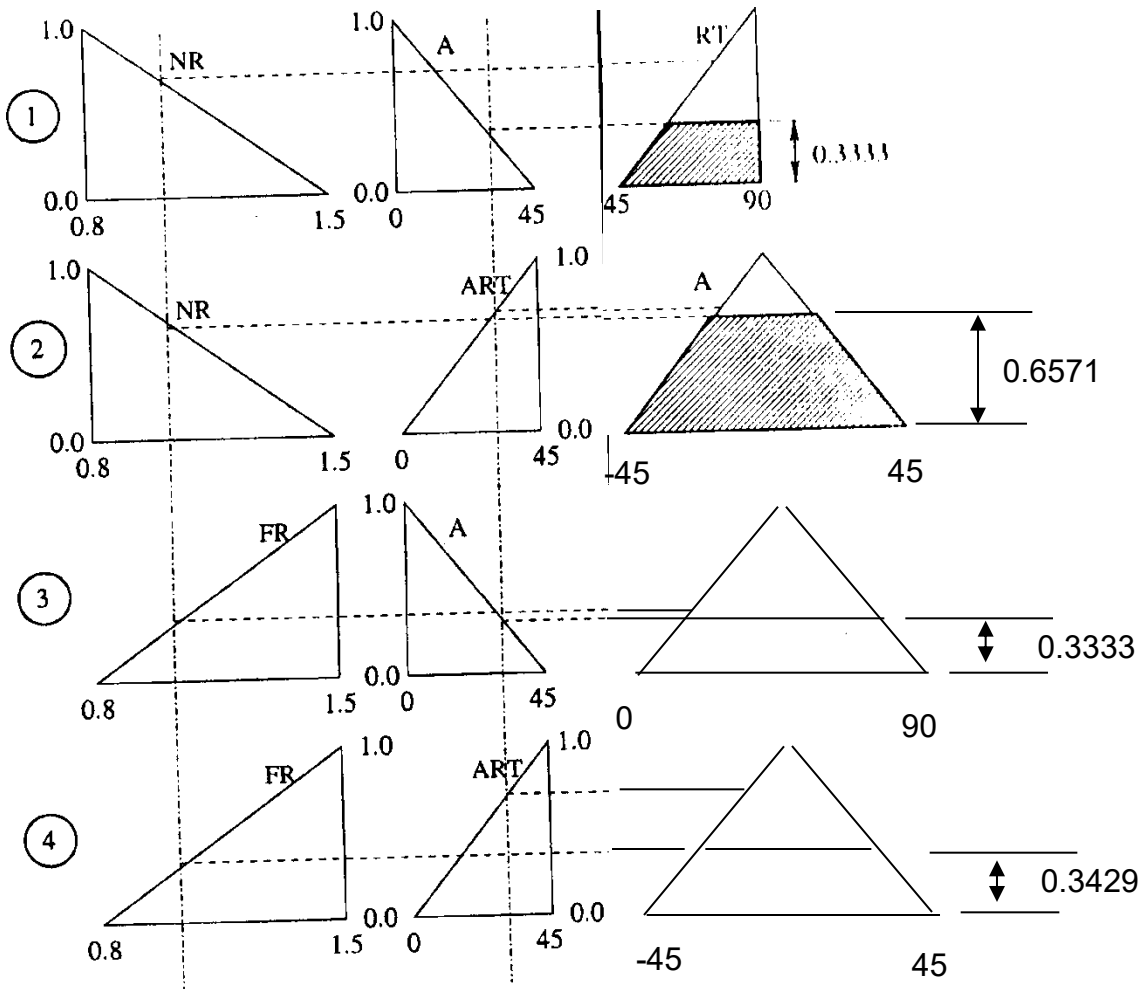


A Numerical Example (Cont.)

- Angle $\text{GSO}_2=30$ degrees declared either A with a membership value of 0.3333 (that is, $\mu_A = 0.3333$) or ART with a membership value of 0.6667 (that is, $\mu_{ART} = 0.6667$).
- For the above set of inputs, the following four rules are being fired from total of 20:
 - If Distance is NR AND Angle is A Then Deviation is RT
 - If Distance is NR AND Angle is ART Then Deviation is A
 - If Distance is FR AND Angle is A Then Deviation is ART
 - If Distance is FR AND Angle is ART Then Deviation is A
- The strengths (α values) of the fired rules are
 - $\alpha_1 = \min(\mu_{NR}, \mu_A) = \min(0.6571, 0.3333) = 0.3333$
 - $\alpha_2 = \min(\mu_{NR}, \mu_{ART}) = \min(0.6571, 0.6667) = 0.6571$
 - $\alpha_3 = \min(\mu_{FR}, \mu_A) = \min(0.3429, 0.3333) = 0.3333$
 - $\alpha_4 = \min(\mu_{FR}, \mu_{ART}) = \min(0.3429, 0.6667) = 0.3429$



A Numerical Example (Cont.)



Distance
input
1.04 m

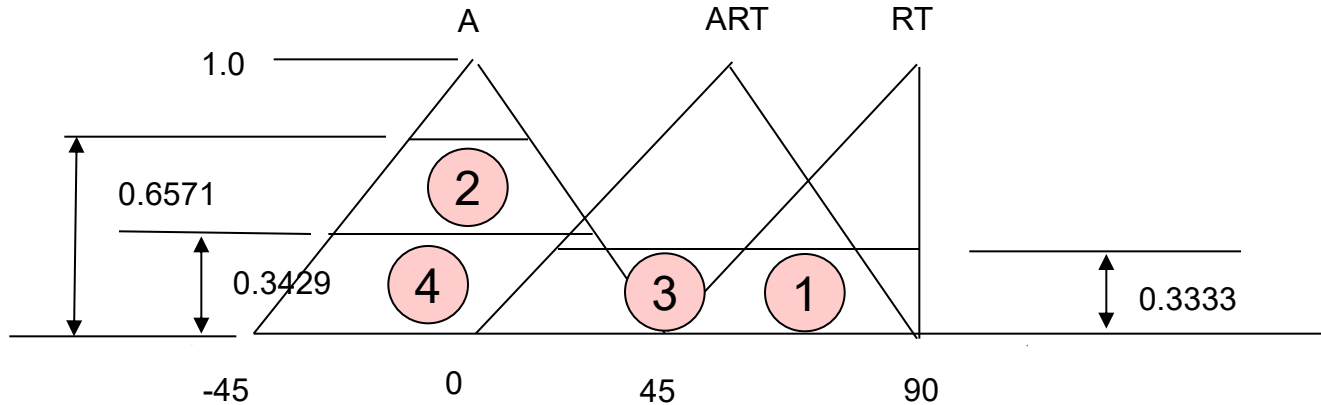
Angle input
30 degrees

Deviation
angle
(degrees)



A Numerical Example (Cont.)

Combined fuzzified output considering all four fired rules.

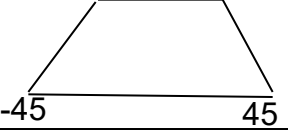
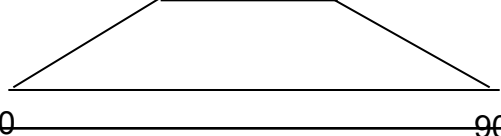
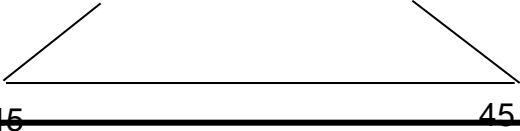




A Numerical Example (Cont.)

Methods of defuzzification

1. Center of Sums Method

Fired rule	Shaded region	Area	Center of area
1		12.5	71
2		39.7089	0
3		25	45
4		25.5699	0



A Numerical Example (Cont.)

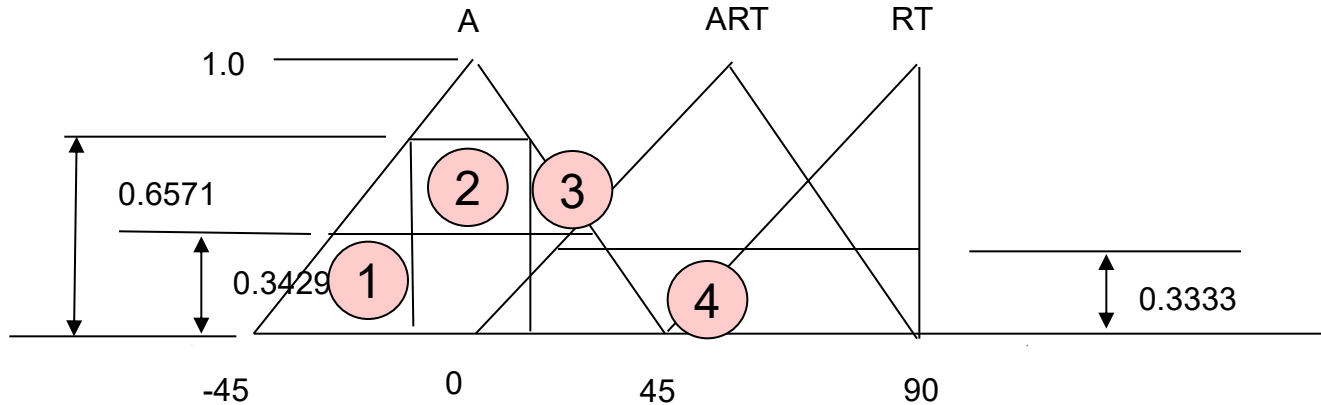
$$U = \frac{12.5 \times 71 + 39.7089 \times 0.0 + 25.0 \times 45.0 + 25.5699 \times 0.0}{12.5 + 39.7089 + 25.0 + 25.5699}$$
$$= 19.5809$$

- Therefore, the robot should deviate by 19.5809 degrees towards right with respect to the line joining the present position of the robot and the goal to avoid collision with the obstacle.



A Numerical Example (Cont.)

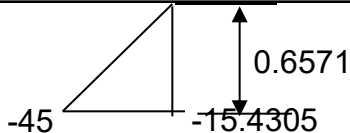
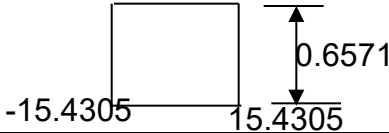
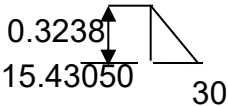
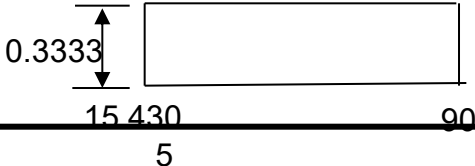
Centroid Method





A Numerical Example (Cont.)

Centroid Method

Marked regions	Shaded region	Area	Center of area
1		9.7151	-25.2860
2		39.7089	0
3		2.3588	20.2870
4		24.8540	52.7153



A Numerical Example (Cont.)

Centroid Method

$$U = \frac{9.7151 \times (-5.2860) + 20.2788 \times 0.0 + 2.3588 \times 20.2870 + 24.8540 \times 52.7153}{9.7151 + 20.2788 + 2.3588 + 24.8540}$$
$$= 19.4450$$



A Numerical Example (Cont.)

- Mean of Maxima: The maximum value of membership (that is, 0.6571) has occurred in a range of deviation angle starting from -15.4305 to 15.4305 degrees.
- Mean = 0.0
- The robot will move along the line joining its present position and the goal.

Takagi and Sugeno's Approach



- A rule is composed of fuzzy antecedent and functional consequent parts
- i-th rule can be represented as follows:
If x_1 is A_1^i and x_2 is A_2^i and x_n is A_n^i then
 $y^i = a_0^i + a_1^i x_1 + \dots + a_n^i x_n$
where $a_0, a_1, \dots, a_n$ are the coefficients
- A non-linear system is considered to be a combination of several linear systems
- Weight of i-th rule can be determined as follows:
 $w^i = \mu_{A_1^i}(x_1) \mu_{A_2^i}(x_2) \dots \mu_{A_n^i}(x_n)$
- Combined control action can be determined as follows:

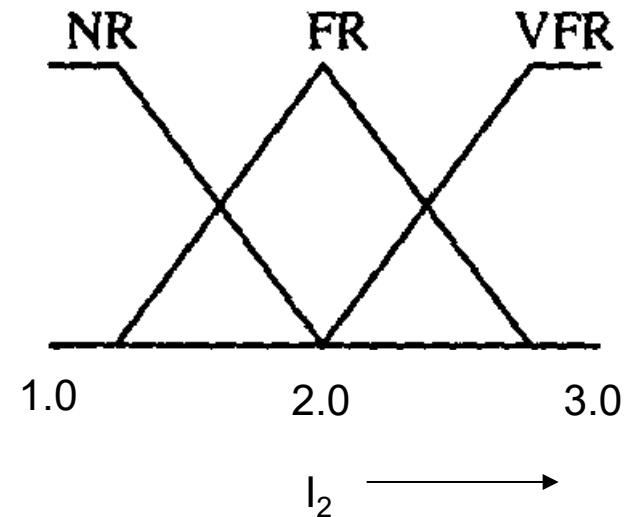
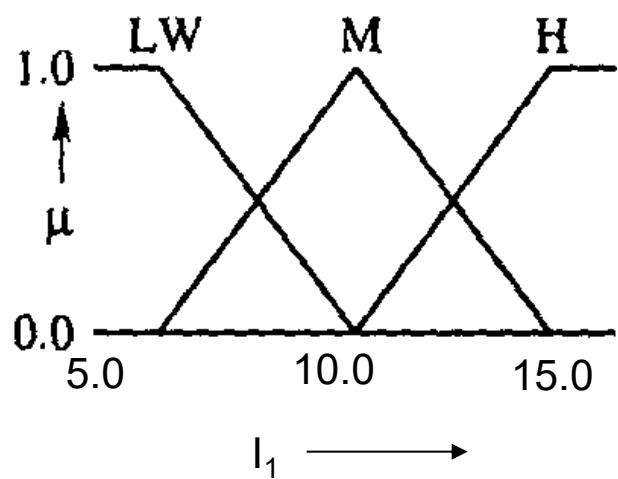
$$y = \frac{\sum_{i=1}^k w^i y^i}{\sum_{i=1}^k w^i}$$

where k indicates the total number of rules.



A Numerical Example

- The DB of the FLC is shown in the figure



- LW: Low
- M: Medium
- H: High

- NR: Near
- FR: Far
- VFR: Very Far



A Numerical Example (Cont.)

- There is a maximum of $3 \times 3 = 9$ feasible rules
- The output of i -th rule, that is, y^i ($i = 1, 2, \dots, 9$) is expressed as follows:

$$y^i = f(I_1, I_2) = a^i_j I_1 + b^i_k I_2$$

- Where $j, k=1, 2, 3$; $a^i_1=1$, $a^i_2=2$ and $a^i_3=3$ if I_1 is found to be LW, M and H respectively. $b^i_1=1$, $b^i_2=2$ and $b^i_3=3$ if I_2 is found to be NR, FR and VFR respectively.
- Inputs: $I_1 = 6.0$, $I_2 = 2.2$.
- I_1 of 6.0 unit can be called either LW (Low) or M (Medium). Similarly, the second input, I_2 of 2.2 unit may be declared either FR (Far) or VFR (Very Far).
- For $I_1 = 6.0$ $\mu_{LW} = 0.8$ and $\mu_M = 0.2$
- For $I_2 = 2.2$. $\mu_{FR} = 0.8$ and $\mu_{VFR} = 0.2$



A Numerical Example (Cont.)

- For the above set of inputs, the following four combinations of input variables lit, fired from a total of nine.
 - I_1 is LW and I_2 is FR,
 - I_1 is LW and I_2 is VFR,
 - I_1 is M and I_2 is FR,
 - I_1 is M and I_2 is VFR.
- weights
 - $w^1 = \mu_{LW} \times \mu_{FR} = 0.8 \times 0.8 = 0.64$
 - $w^2 = \mu_{LW} \times \mu_{VFR} = 0.8 \times 0.2 = 0.16$
 - $w^3 = \mu_M \times \mu_{FR} = 0.2 \times 0.8 = 0.16$
 - $w^4 = \mu_M \times \mu_{VFR} = 0.2 \times 0.2 = 0.04$



A Numerical Example (Cont.)

- Functional consequent values

- $y^1 = I_1 + 2I_2 = 6.0 + 2 \times 2.2 = 10.4$
- $y^2 = I_1 + 3I_2 = 6.0 + 3 \times 2.2 = 12.6$
- $y^3 = 2I_1 + 2I_2 = 2 \times 6.0 + 2 \times 2.2 = 16.4$
- $y^4 = 2I_1 + 3I_2 = 2 \times 6.0 + 3 \times 2.2 = 18.6$

$$w^1 y^1 + w^2 y^2 + w^3 y^3 + w^4 y^4$$

- $$y = \frac{w^1 y^1 + w^2 y^2 + w^3 y^3 + w^4 y^4}{w^1 + w^2 + w^3 + w^4}$$

$$= 12.04$$

Advantages and Disadvantages of Fuzzy Logic Controller



- Advantages
 - Potential tool for dealing with imprecision and uncertainty
 - Does not require an extensive mathematical formulation of the problem
- Disadvantages
 - Designing a proper KB of an FLC is a difficult task
 - May not be suitable for modeling a process involving many variables



References

1. E. H. Mamdani, S. Assilian, “An experiment in linguistic synthesis with fuzzy logic controller,” *International Journal of Man-Machine Studies*, 7, pp. 1-13, 1975.
2. T. Takagi, M. Sugeno,” Fuzzy identification of systems and its application to modeling and control,” *IEEE Trans. On Systems, Man and Cybernetics*, SMC-15, pp. 116-132, 1985.
3. J.C. Bezdek, “Fuzzy mathematics in pattern classification,” *Ph. D. thesis*, Applied Mathematics Center, Cornell University, Ithaca, 1973.
4. J. Yao, M. Dash, S.T. Tan, H. Liu, “Entropy-based fuzzy clustering and fuzzy modeling,” *Fuzzy Sets and Systems*, 113, pp. 381-388, 2000.



Thank you

Questions?